

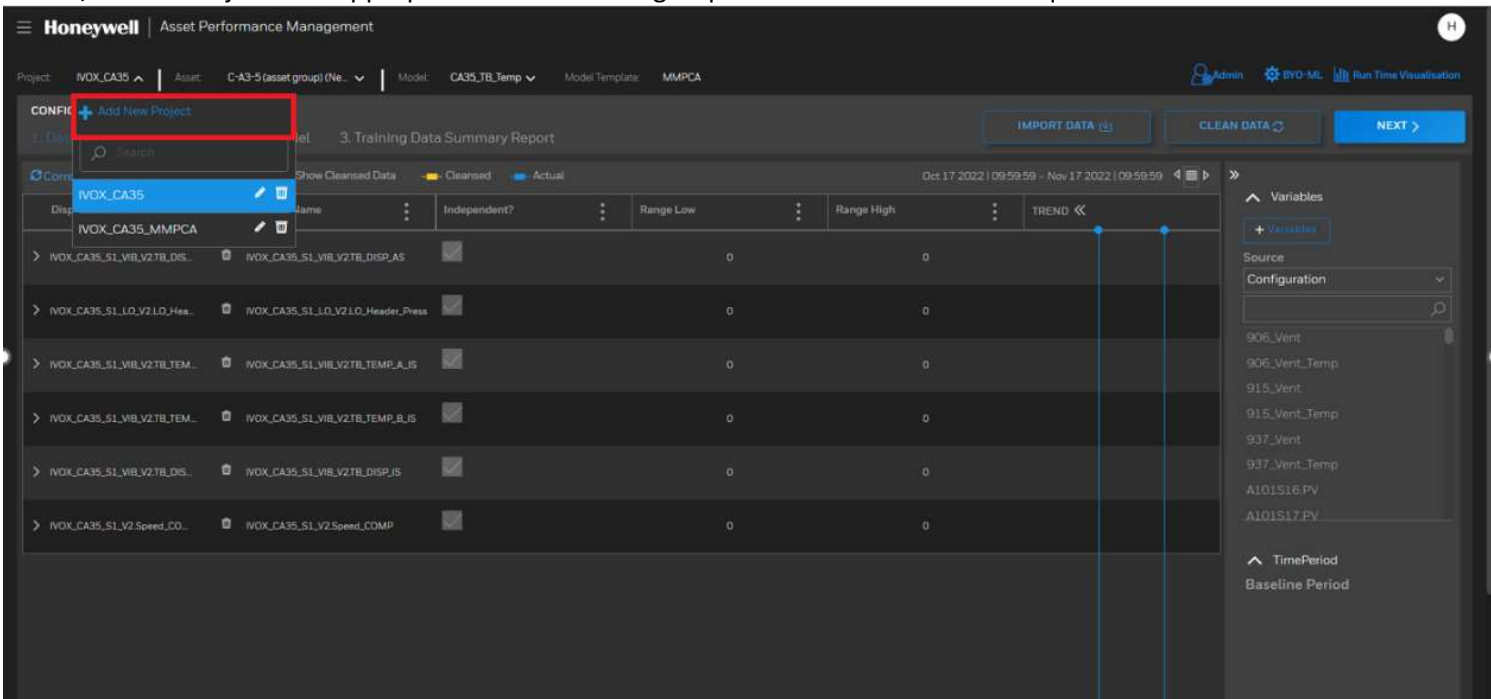
MMPCA Model Development

Prechecks before creating model

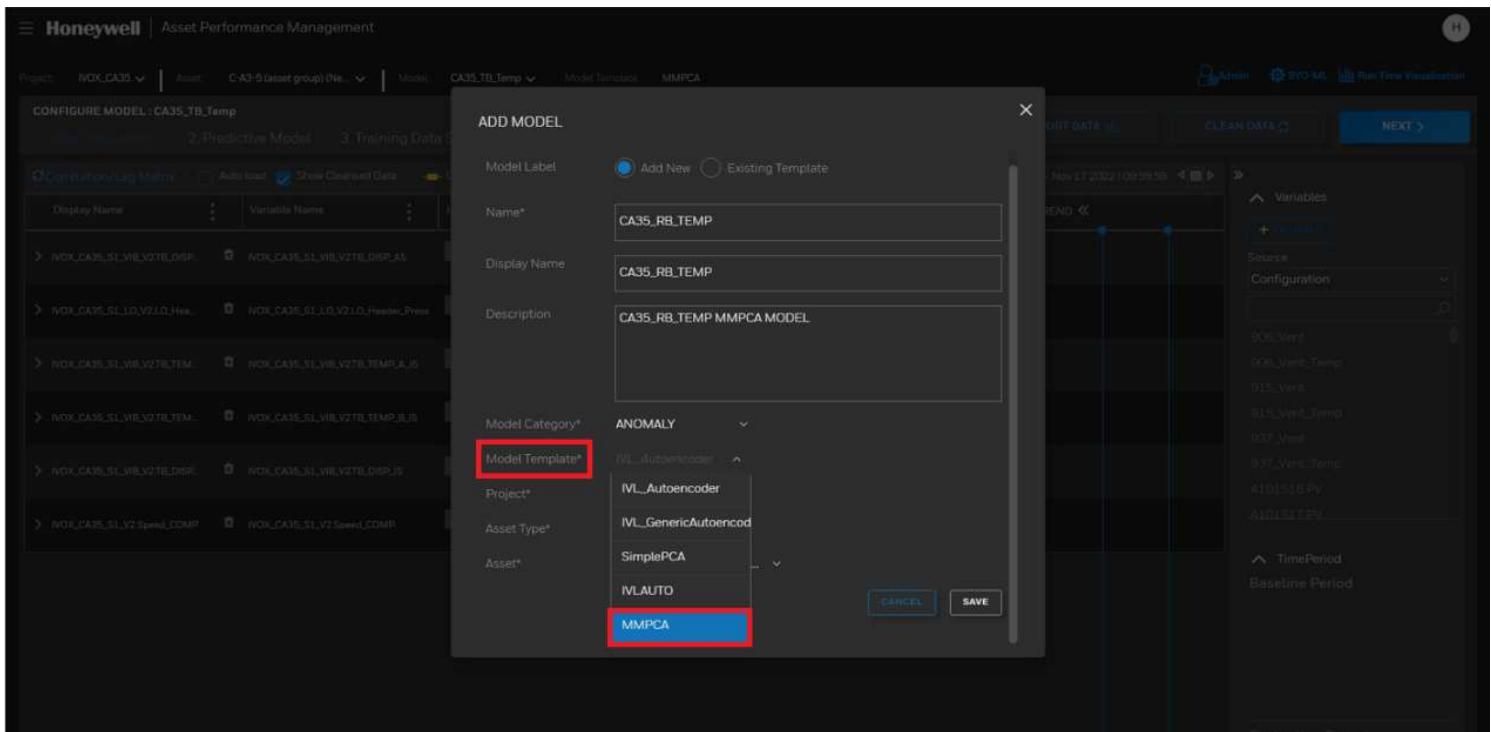
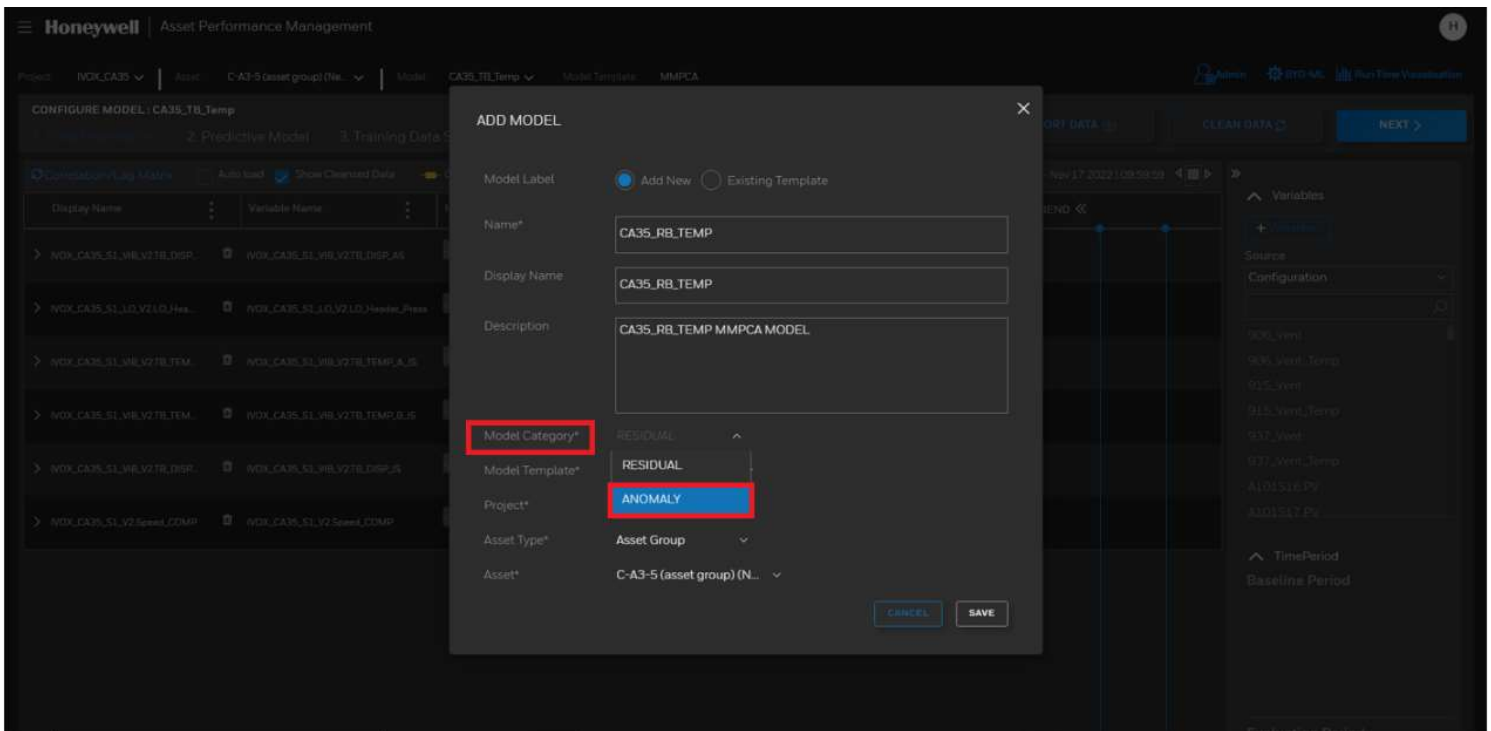
- 1. UOM for all the attributes must be in Type Level (SI Unit) in “Asset Type Attributes” sheet
- 2. UOM for all the attributes must be in Instance level in the mapping sheets (“Asset Attribute Mapping”, “Attribute Property Mapping”)

MMPCA Model Building Procedure

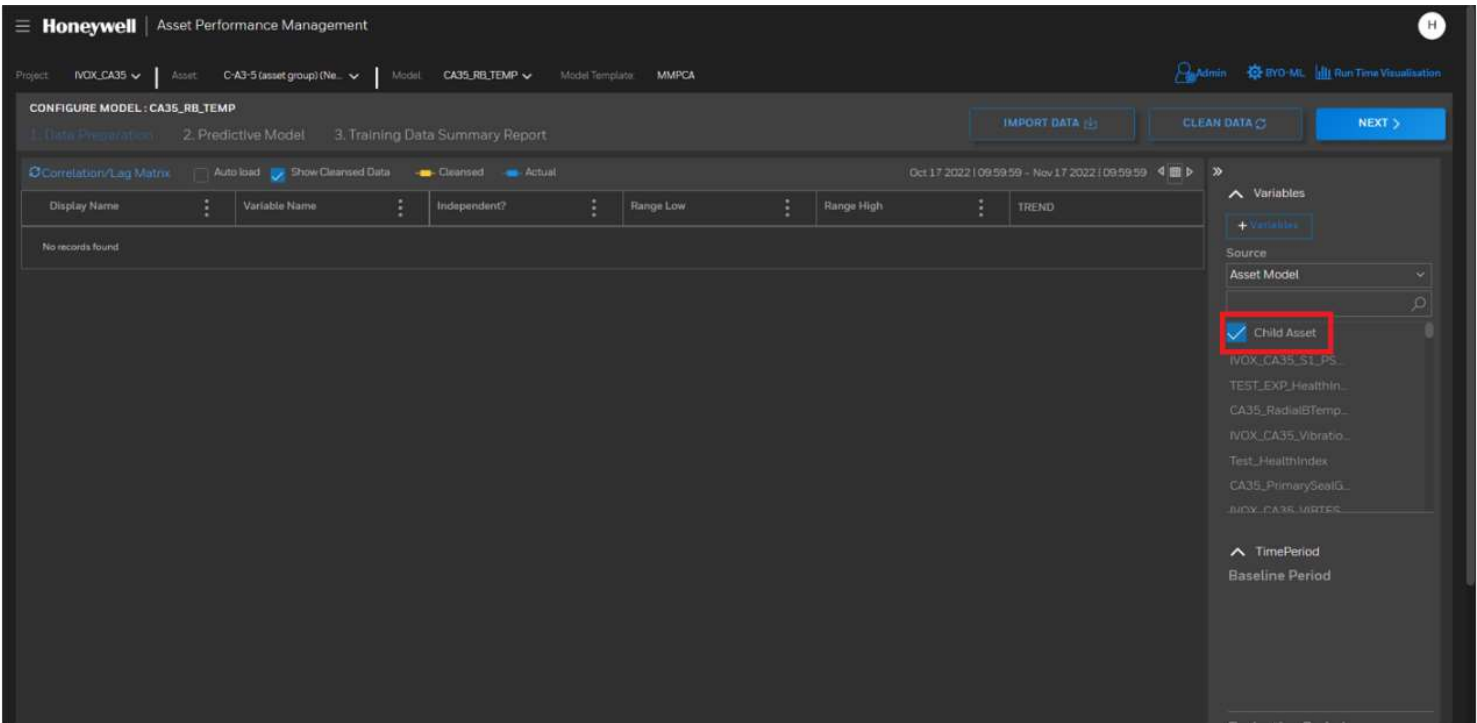
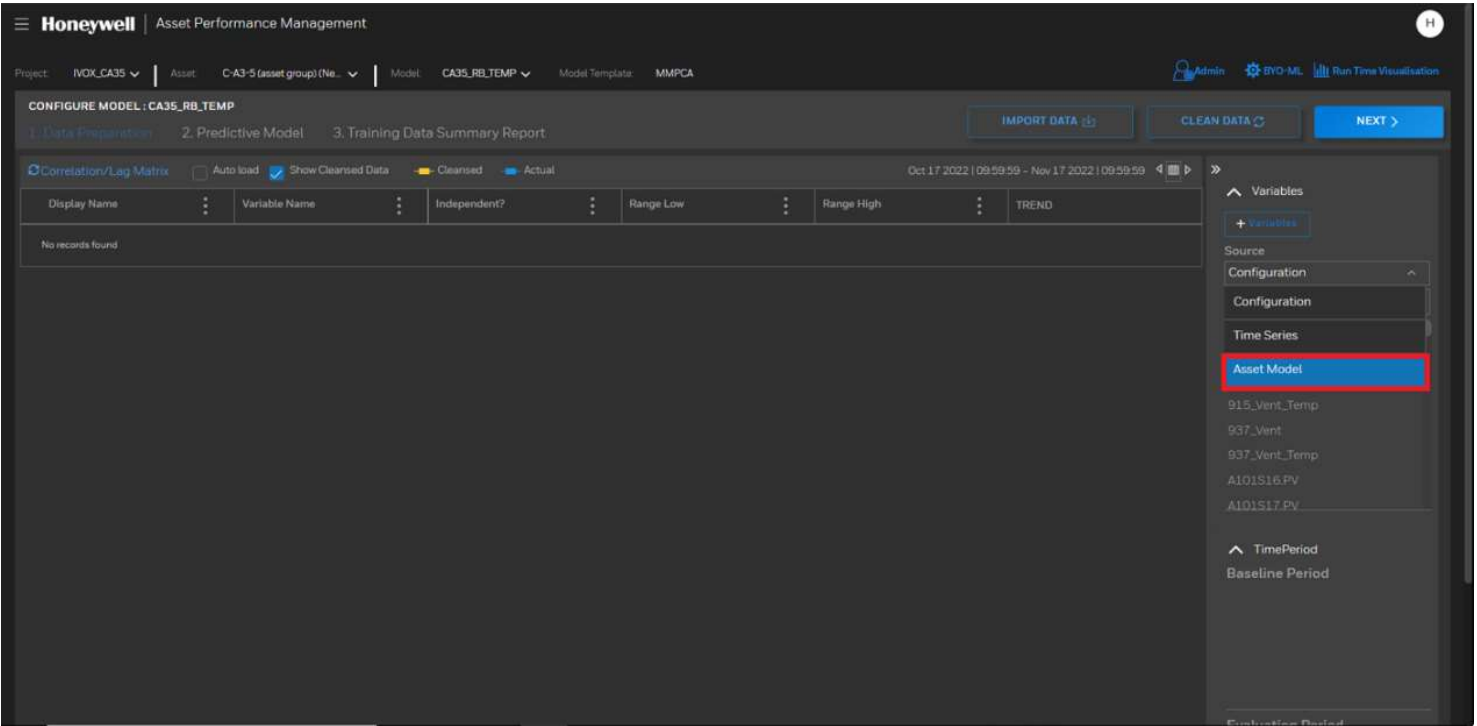
- 1. Select / Create Project with appropriate asset or asset group based on the attributes required in the model



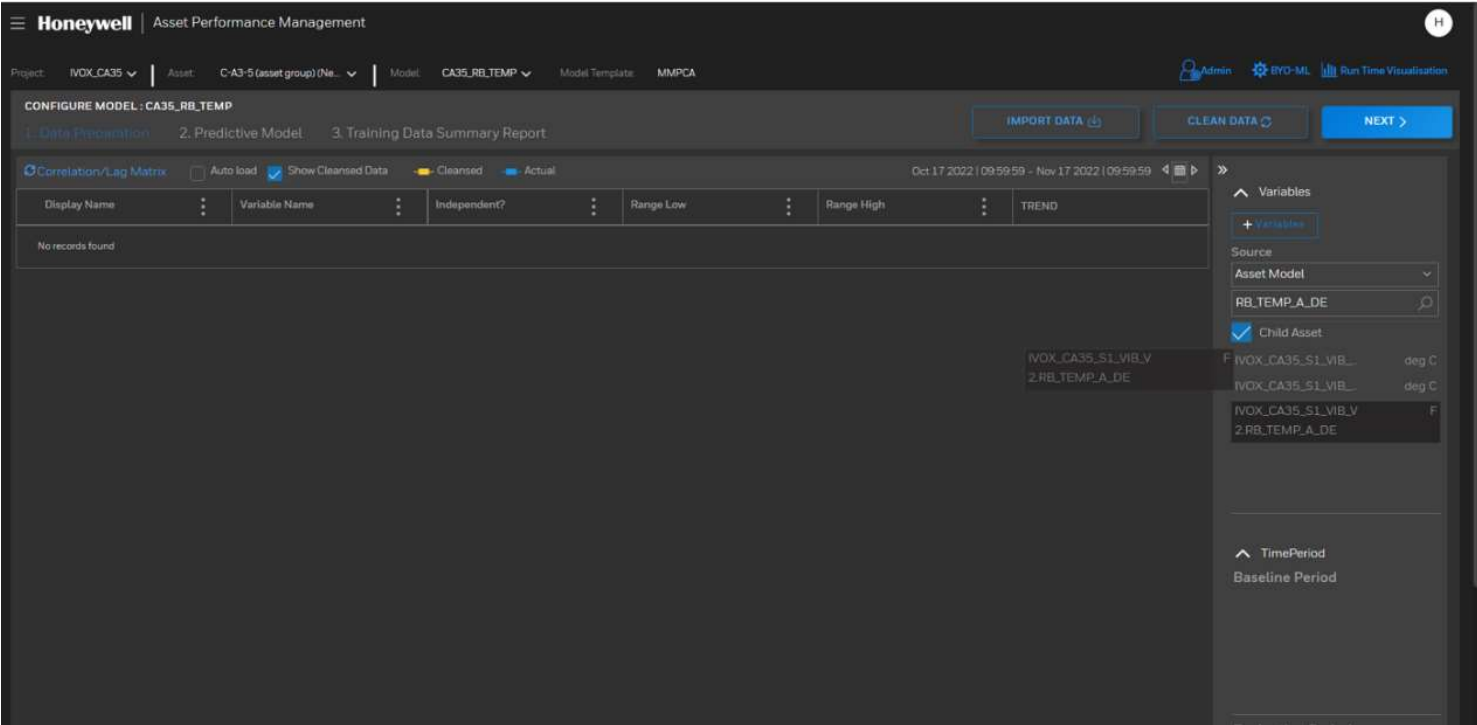
- 2. Create new model with Model Category as “Anomaly” and Model template as “MMPCA”



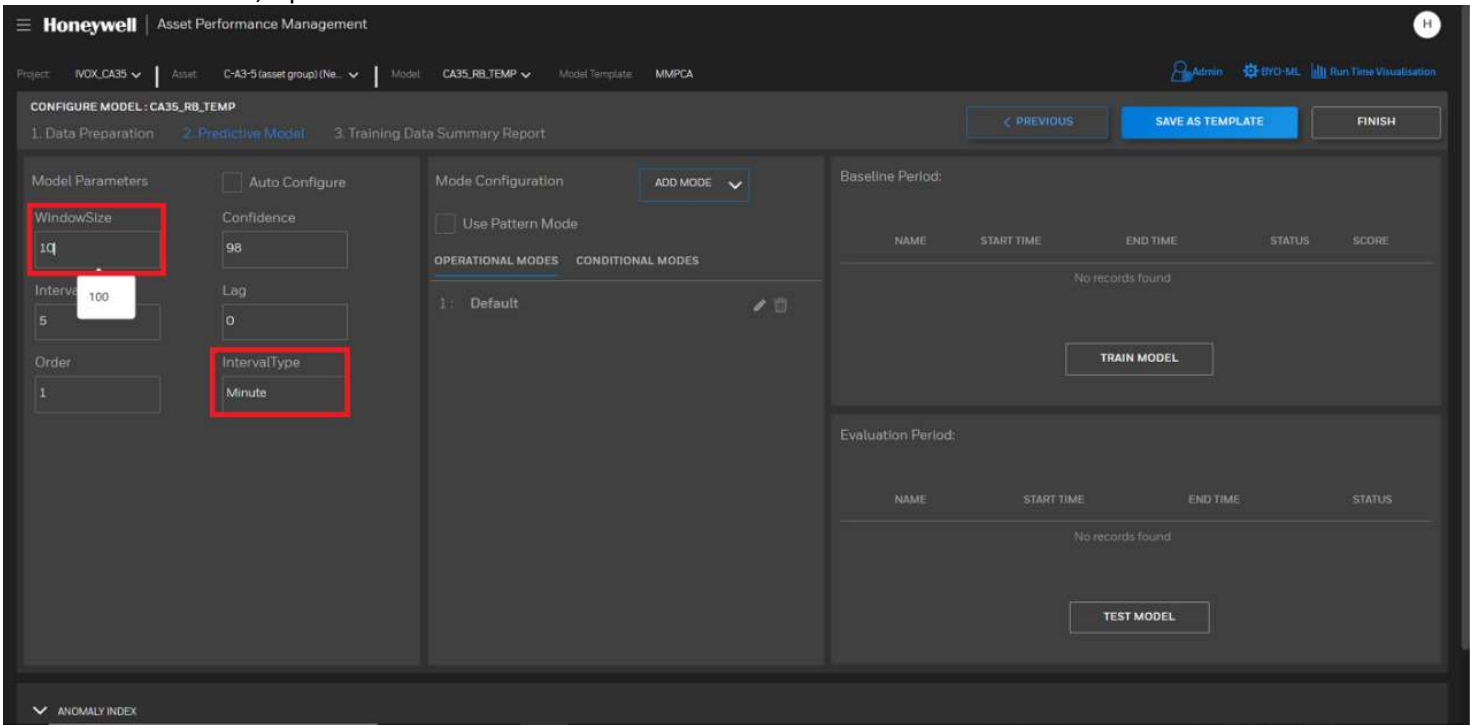
3. To add new attributes use source as “Asset Model” and include the child asset



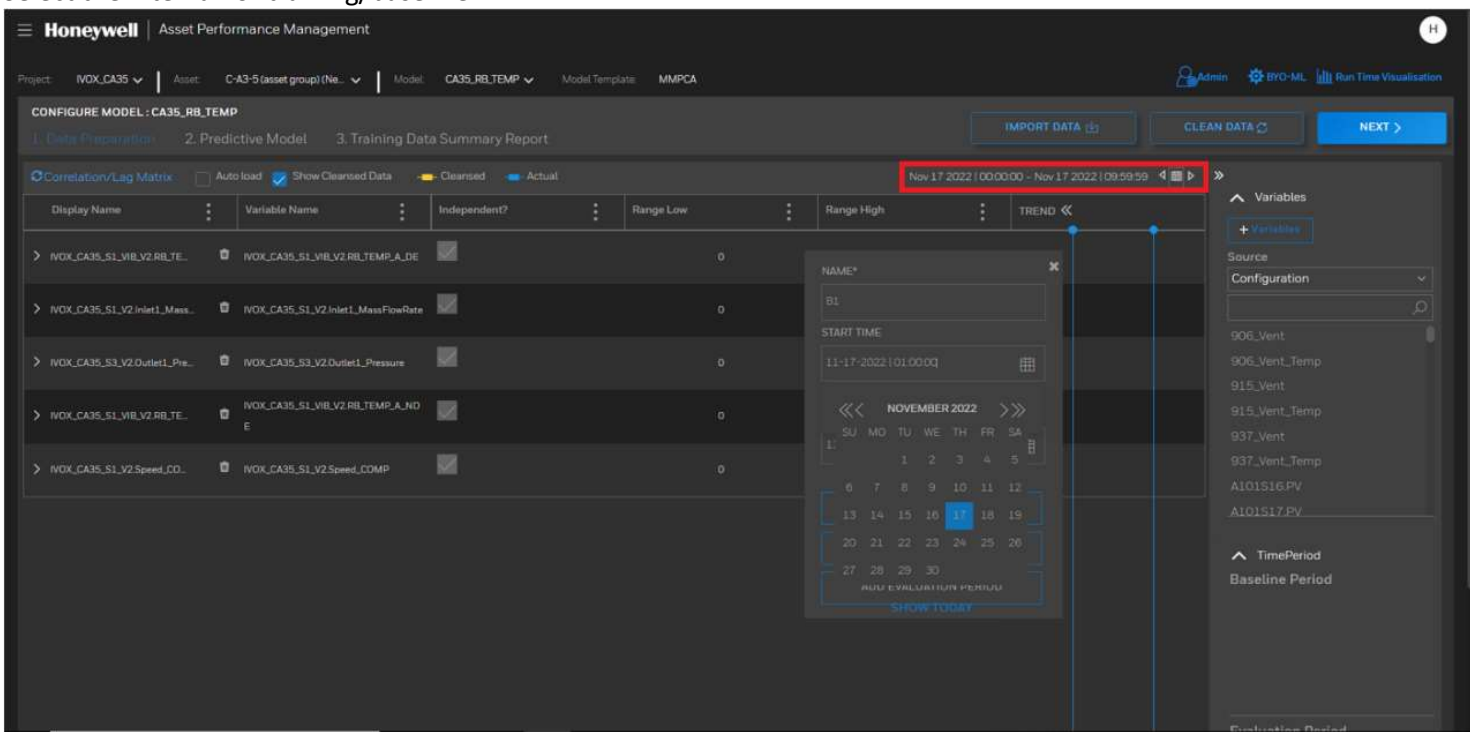
4. Search for required attributes, drag and drop on the dashboard



5. Go to “NEXT” section, Update the Window size and Interval

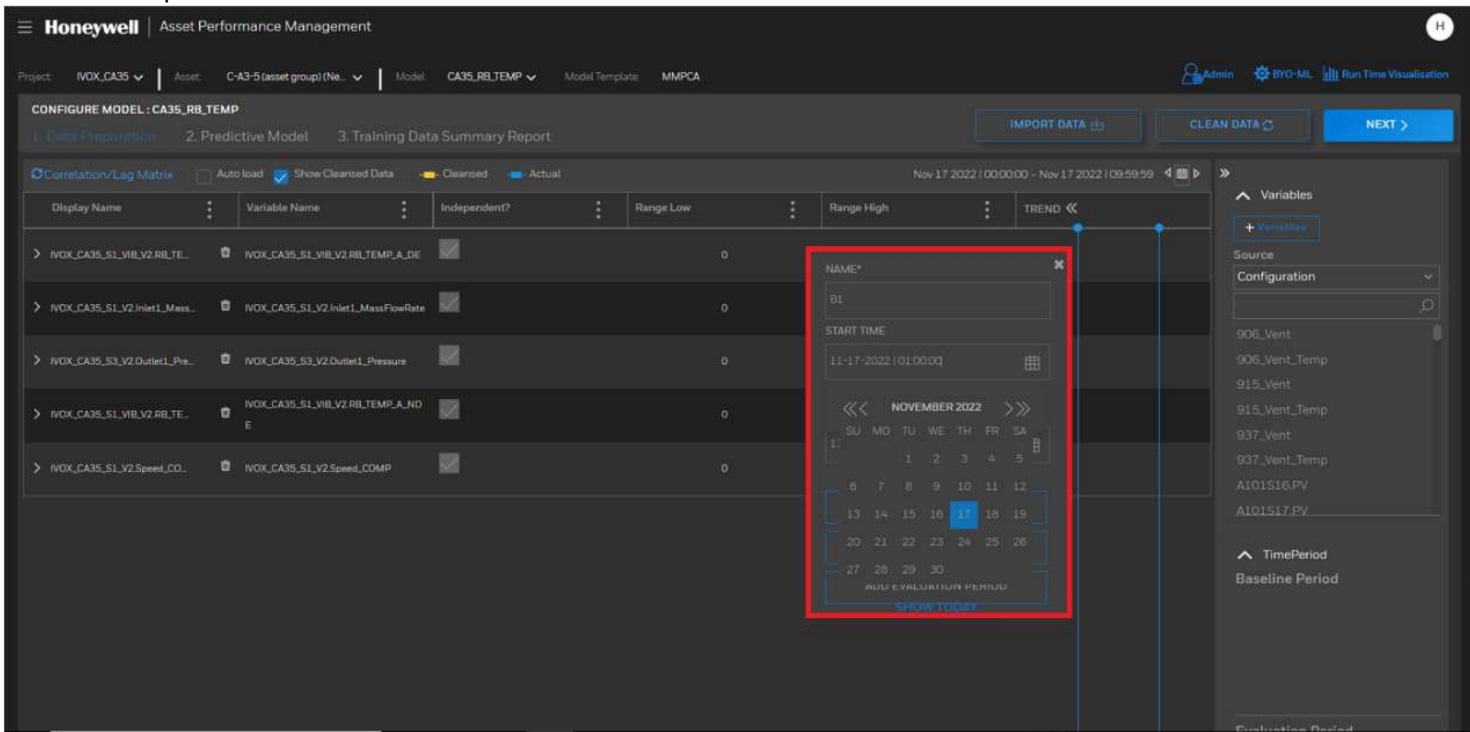


6. Select the interval for training/baseline



7. Manually clean the data using the “Clean Data” button for baseline intervals

8. Add baseline period



9. Go to Next Section and train data

Project: IVOX_CA35

Asset: C-A3-5 (asset group) (Ne...

Model: CA35_RB_TEMP

Model Template: MMPCA

Admin

BYO-ML

Run Time Visualisation

CONFIGURE MODEL: CA35_RB_TEMP

1. Data Preparation

2. Predictive Model

3. Training Data Summary Report

PREVIOUS

SAVE AS TEMPLATE

FINISH

Model Parameters

☐ Auto Configure

WindowSize: 100

Confidence: 98

Interval: 1

Lag: 0

Order: 1

IntervalType: Minute

Mode Configuration

ADD MODE

☐ Use Pattern Mode

OPERATIONAL MODES

1: Default

Baseline Period:

☒ NAME

☒ B3

☒ B5

START TIME	END TIME	STATUS	SCORE
11-25-2019 12:00:00 AM	11-30-2019 12:00:00 AM		0
01-25-2020 12:00:00 AM	01-30-2020 12:00:00 AM		0
05-01-2019 12:00:00 AM	05-10-2019 12:00:00 AM		

TRAIN MODEL

Evaluation Period:

NAME	START TIME	END TIME	STATUS
No records found			

TEST MODEL

ANOMALY INDEX

Last 1 Week

CONFIGURE LIMIT

Project: IVOX_CA35

Asset: C-A3-5 (asset group) (Ne...

Model: CA35_TB_Temp

Model Template: MMPCA

Admin

BYO-ML

Run Time Visualisation

CONFIGURE MODEL: CA35_TB_Temp

1. Data Preparation

2. Predictive Model

3. Training Data Summary Report

PREVIOUS

SAVE AS TEMPLATE

FINISH

Model Parameters

☐ Auto Configure

WindowSize: 100

Confidence: 98

Interval: 1

Lag: 0

Order: 1

IntervalType: Minute

Mode Configuration

ADD MODE

☐ Use Pattern Mode

OPERATIONAL MODES

1: Default

Baseline Period:

☒ NAME

☒ B1

START TIME	END TIME	STATUS	SCORE
05-01-2019 12:00:00 AM	05-10-2019 12:00:00 AM		0
07-01-2019 12:00:00 AM	07-05-2019 12:00:00 AM		0
08-01-2019 12:00:00 AM	08-05-2019 12:00:00 AM		

TRAIN MODEL

Evaluation Period:

NAME	START TIME	END TIME	STATUS
No records found			

TEST MODEL

ANOMALY INDEX

Last 1 Week

CONFIGURE LIMIT

Model Training in progress

* Progress bar will disappear after 2 mins and model training continues in the background, refresh UI to check the status.

CLOSE

10. Once training is completed, check the score of training (score above 90 can be considered good)

Project: IVOX_CA35

Asset: C-A3-5 (asset group) (Ne...

Model: CA35_TB_Temp

Model Template: MMPCA

Admin

BYO-ML

Run Time Visualisation

CONFIGURE MODEL: CA35_TB_Temp

1. Data Preparation

2. Predictive Model

3. Training Data Summary Report

PREVIOUS

SAVE AS TEMPLATE

FINISH

Model Parameters

☐ Auto Configure

WindowSize: 100

Confidence: 98

Interval: 1

Lag: 0

Order: 1

IntervalType: Minute

Mode Configuration

ADD MODE

☐ Use Pattern Mode

OPERATIONAL MODES

1: Default

Baseline Period:

☒ NAME

☒ B1

☒ B2

START TIME	END TIME	STATUS	SCORE
05-01-2019 12:00:00 AM	05-10-2019 12:00:00 AM		100
07-01-2019 12:00:00 AM	07-05-2019 12:00:00 AM		100
08-01-2019 12:00:00 AM	08-05-2019 12:00:00 AM		

RETRAIN MODEL

Evaluation Period:

NAME	START TIME	END TIME	STATUS
No records found			

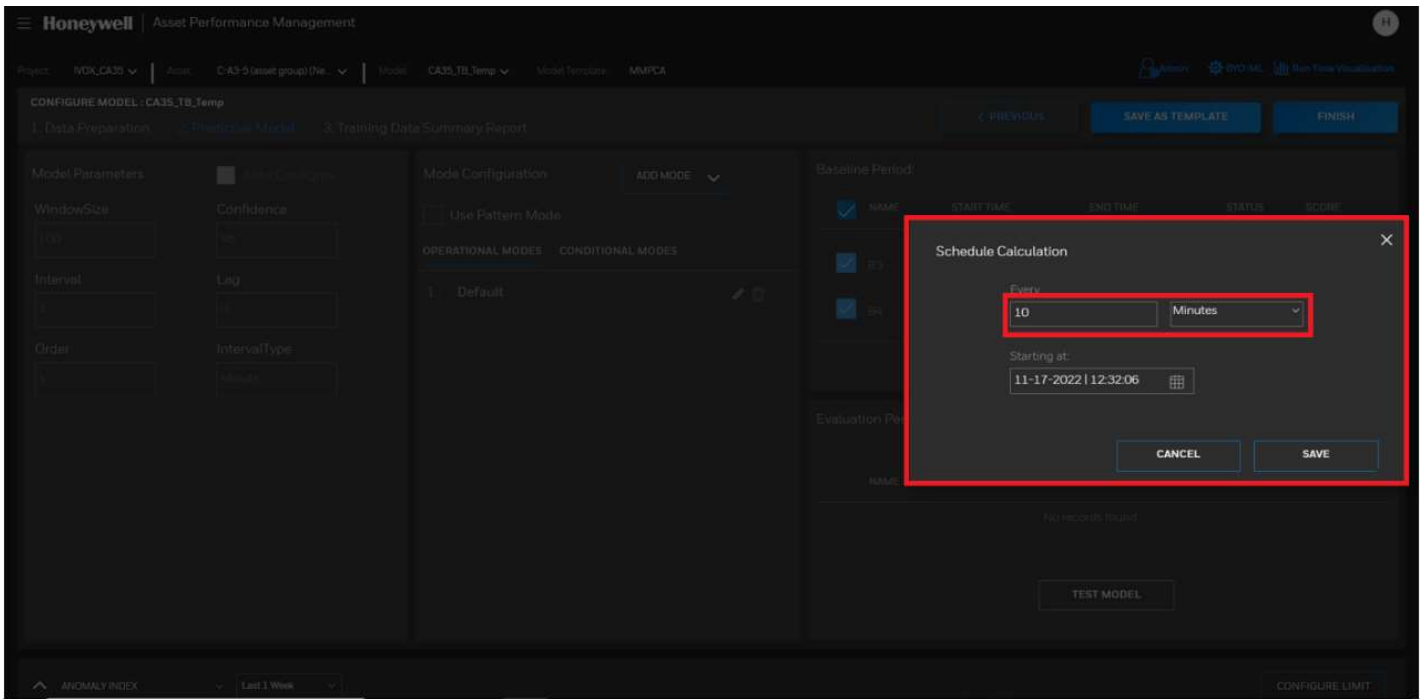
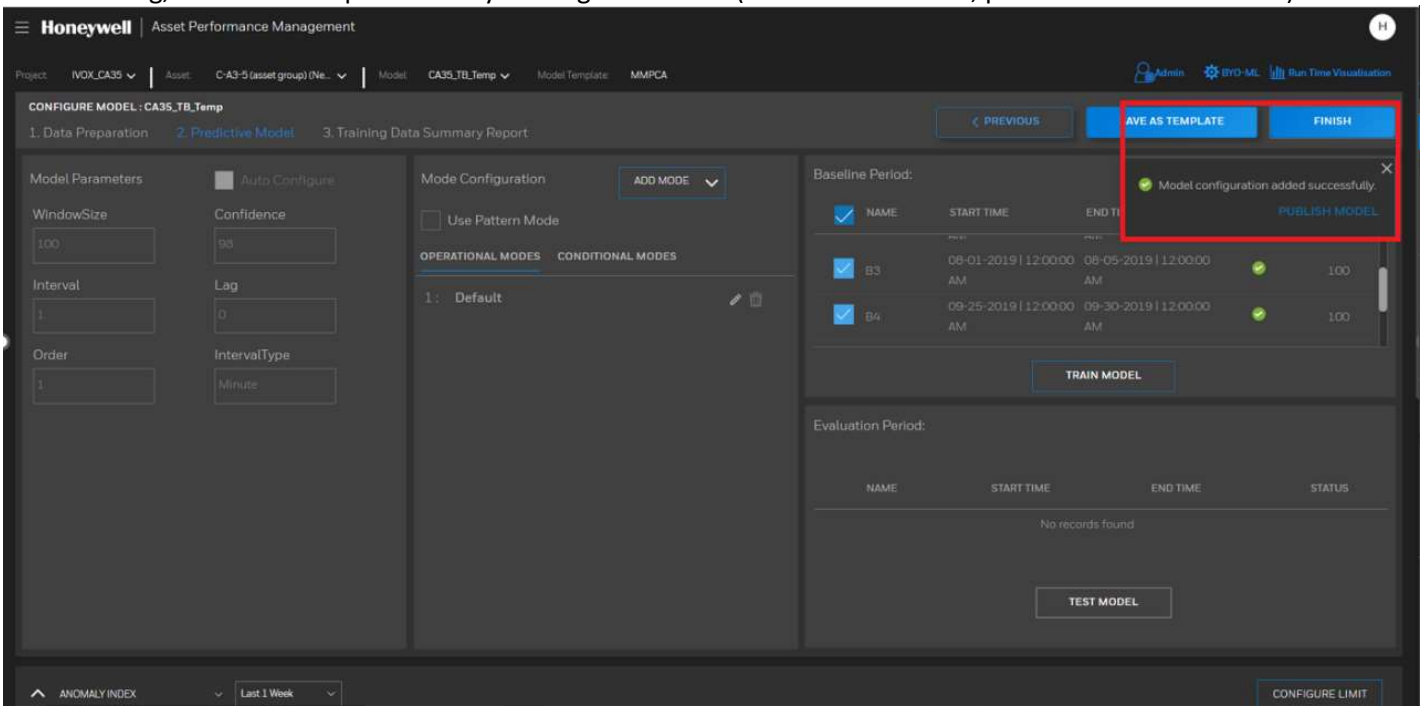
TEST MODEL

ANOMALY INDEX

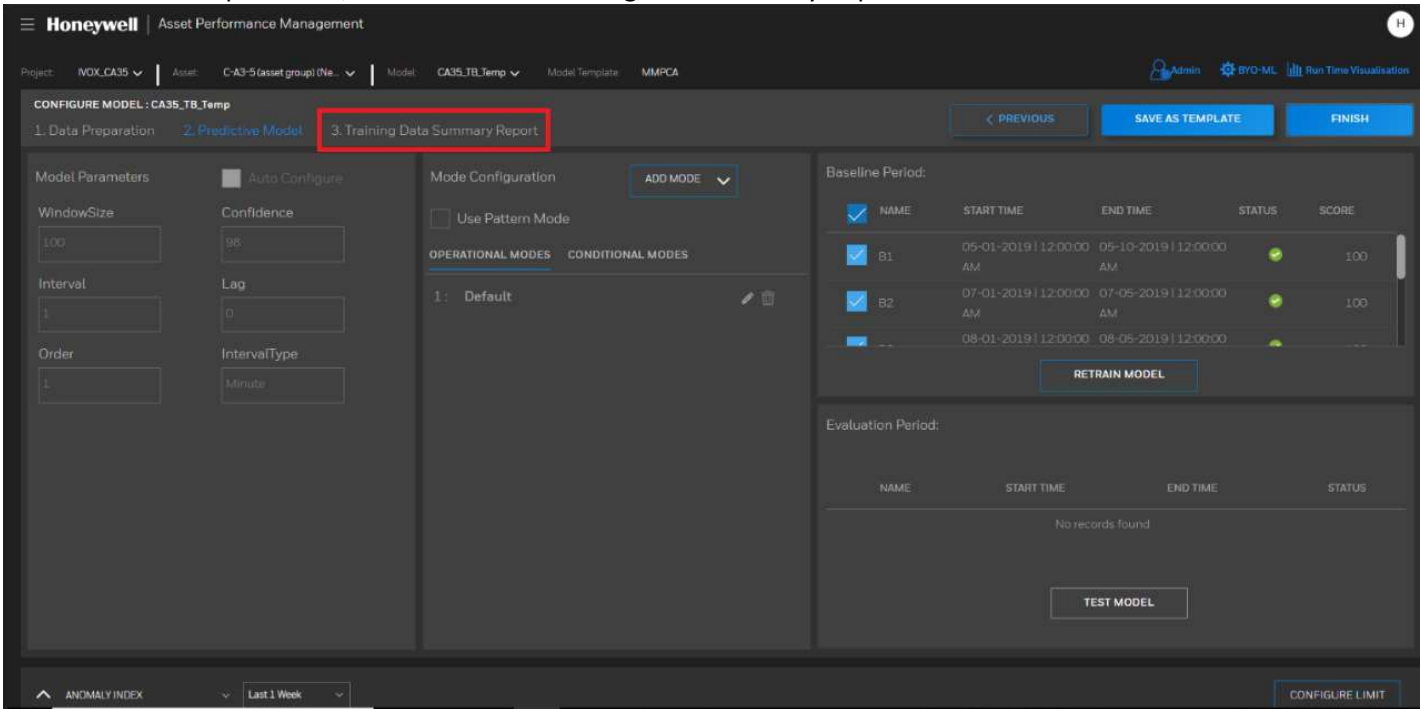
Last 1 Week

CONFIGURE LIMIT

11. After training, model can be published by clicking on “Finish” (Must be scheduled, preferred 10 min interval)



12. Once the model is published, download the “Training Data Summary Report”



13. Mean, Baseline SD must be updated through BCT/Sentinel for the attributes as per the “Training Data Summary Report”
14. Map the expected values through BCT from “Attribute Property Mapping” sheet
15. Enable the faults based on attribute availability, if required faults can be added as per blueprint
16. Add causes, recommendations for the faults that have been enabled

POINTS TO REMEMBER

1. Manual cleaning interval must not be more than one month period. Manual cleaning can be done multiple times over multiple intervals, but care must be taken that the interval selected is not more than one month
2. Window Size and Interval must be given even before manual cleaning and training of data
3. Expected Value tag format to add in BCT: <Asset Group>.<Asset Model>.<Attribute>.<ExpectedValue>